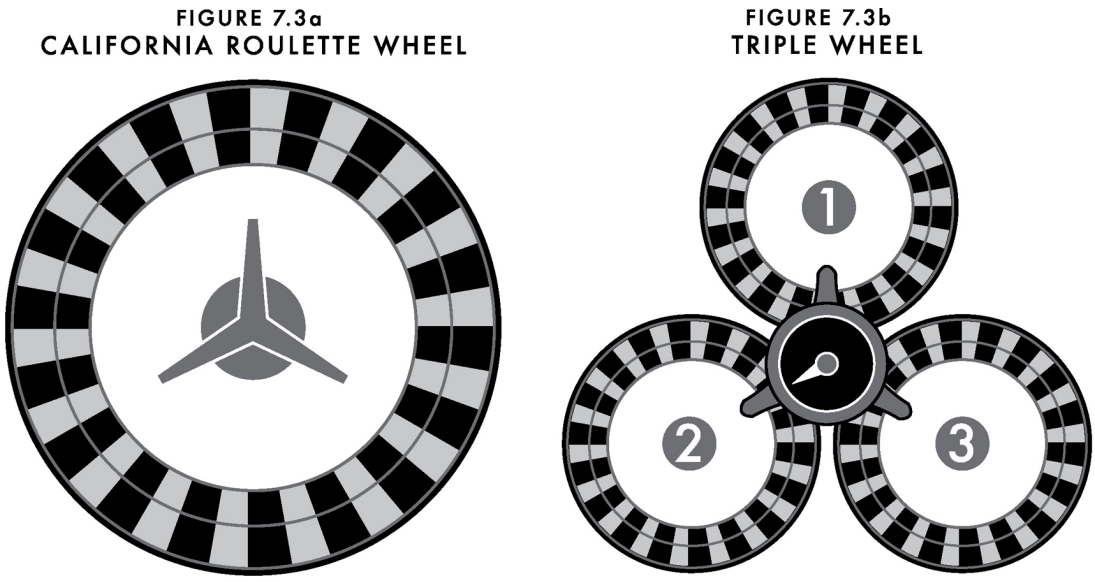


Figure 7.3. (a) A California roulette wheel. (b) The “triple wheel” at Ronnie’s casino, with the selector dial in the center.

Gillian and Fatima do this many times over, to inspect the properties of these wheels thoroughly, and they take detailed notes on the outcomes: which wheel was used and what color came up on each run. Black and red show up in roughly equal proportions, and after several dozen runs, the inspectors go back to their office to compare their notes.

The inspectors find that each table’s roulette wheels really do seem completely random—red and black each came up almost exactly half the time. But there are strange correlations between Fatima’s notes and Gillian’s notes. Each time the small dials at the two tables selected the same wheel number, the two roulette balls landed on the same color. For example, on run 87, both dials pointed to wheel 2—and both roulette balls landed on red (Figure 7.5). The inspectors conclude that the balls are preprogrammed in the giant roulette-ball machine to ensure they always land on the same color when they go to matching wheels.